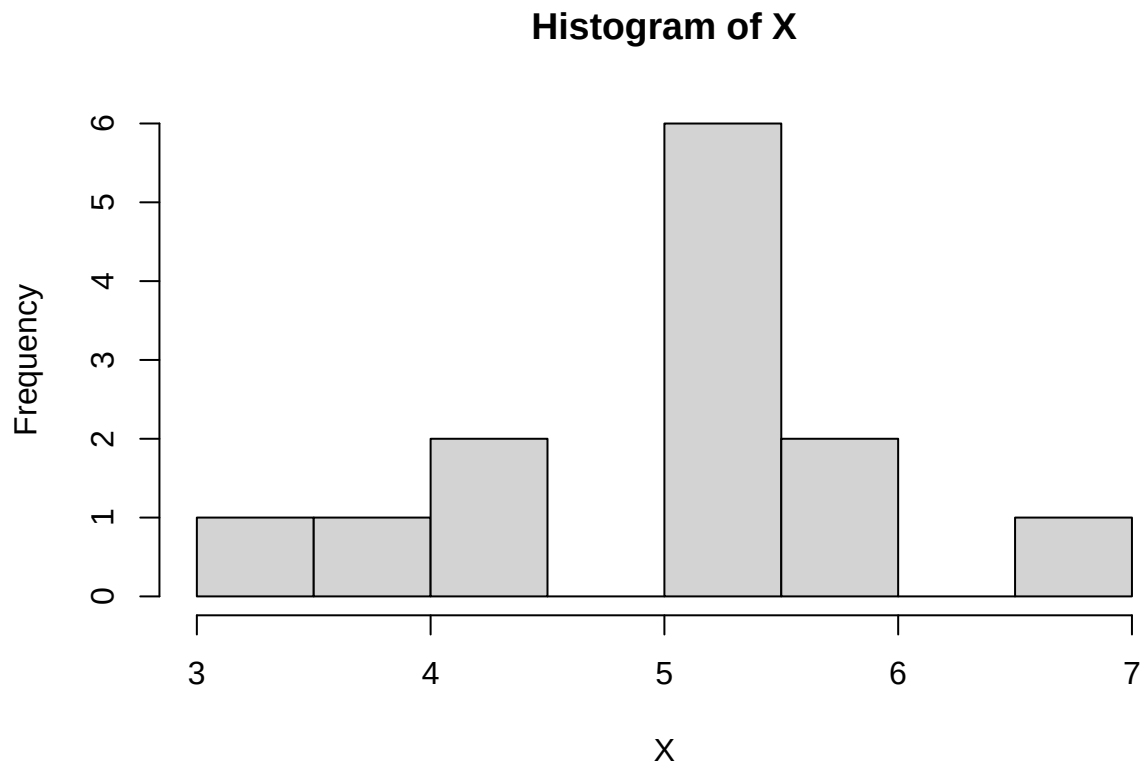


lab5

2026-02-05

Q1

$$F^{-1}(x) = \mu + c - \log(-\log(x))$$

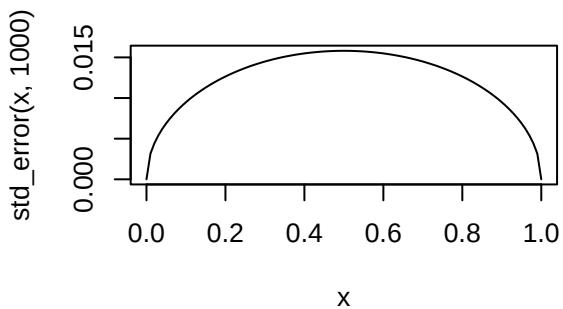
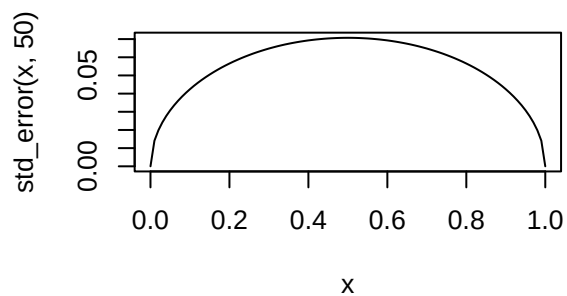
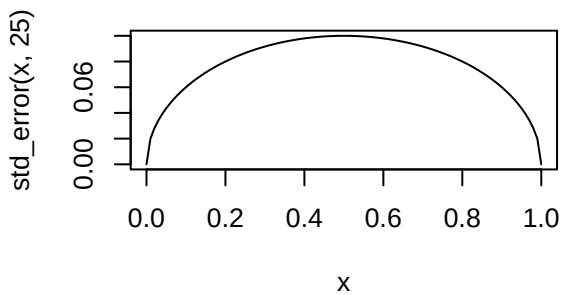
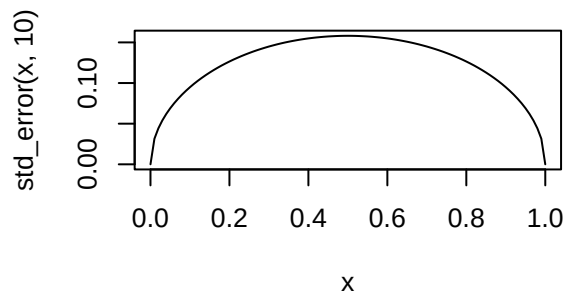


b Say our power for a test for a given mu is p. We estimate p by

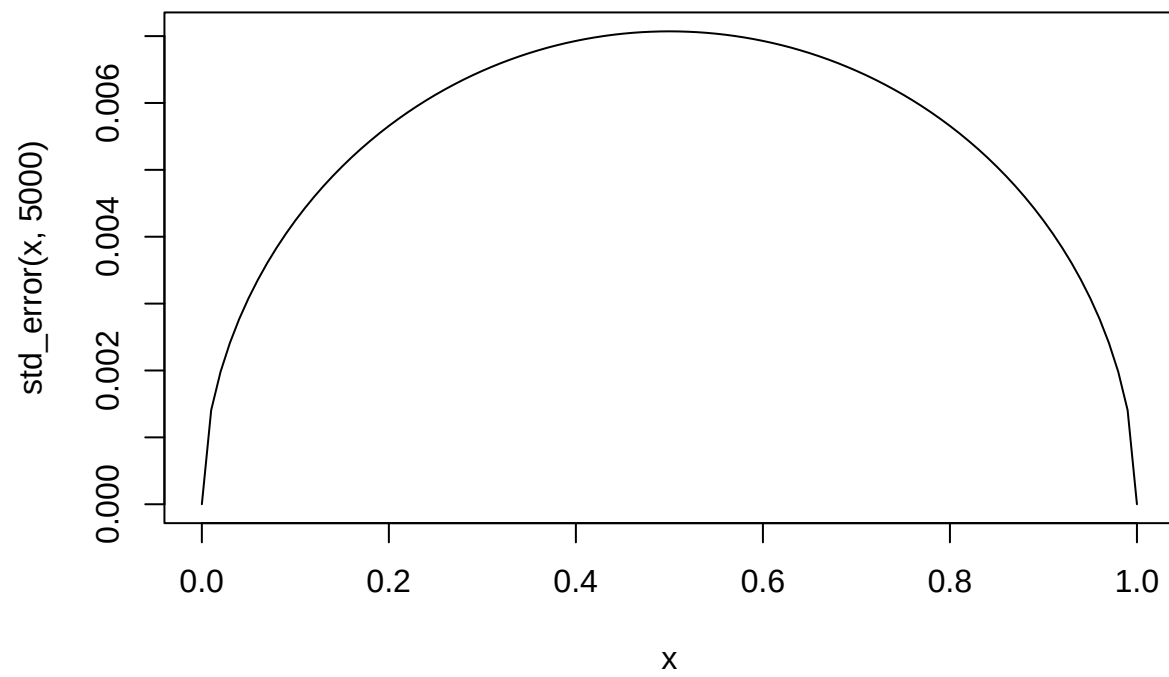
$$\frac{\text{\#number of rejections}}{\text{repetitions}} = \frac{\text{\#number of rejections}}{n} = \hat{p}$$

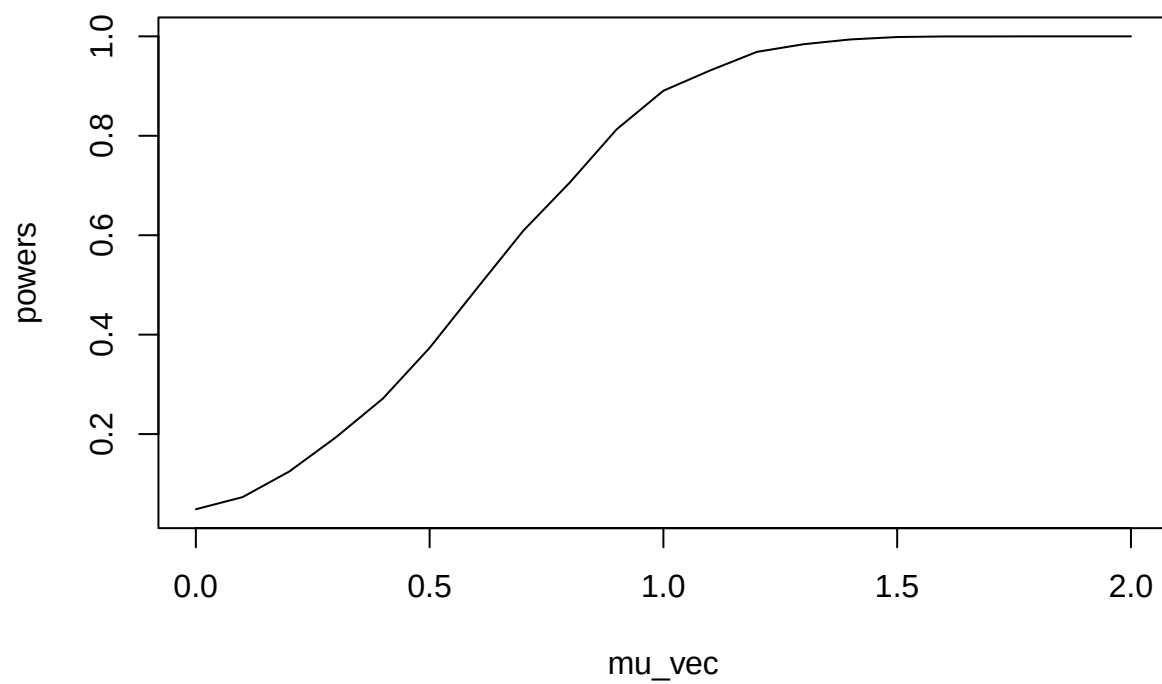
$$\text{\#number of rejections} \sim \text{bin}(n, p), \text{Var}(\hat{p}) = p(1 - p)/n$$

The plots below show $\sqrt{\text{Var}(\hat{p})}$ for different n.



Say we have standard error = $0.015 = \sigma$, then our power estimate varies by roughly $3\sigma = 0.045$. It depends on what precision we are after. As our task is not heavy computationally, lets just increase the repetitions alot, say to 5000.



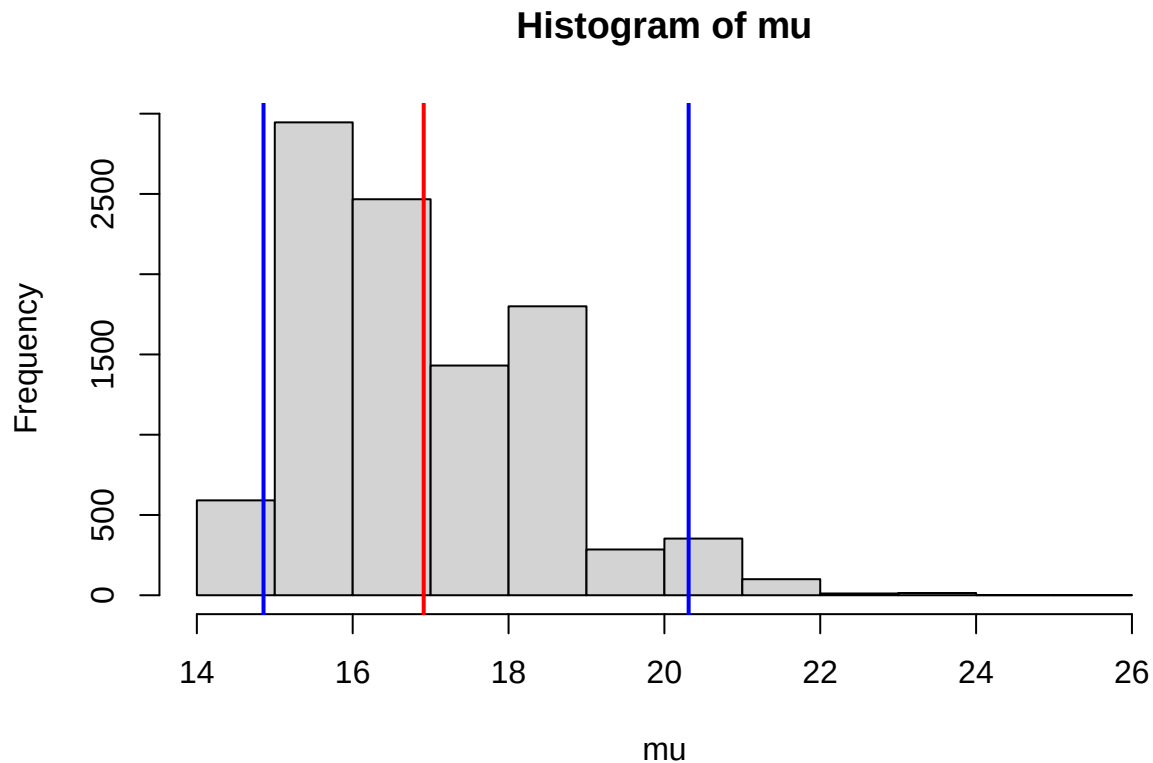


If we don't have variance we expect the power function to be monotonically increasing.

Q2

a

In red we see the mean of our sample, in blue we have the 95% CI.



```
## CI: 14.85556 20.31111
```

b

For the CI we use a standard t-distribution CI as the true variance is unknown.

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 10000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = cb, type = "bca")
##
## Intervals :
## Level      BCa
## 95%      (15.08, 22.53 )
## Calculations and Intervals on Original Scale

## [1] 13.15397 20.66825
## attr(,"conf.level")
## [1] 0.95
```

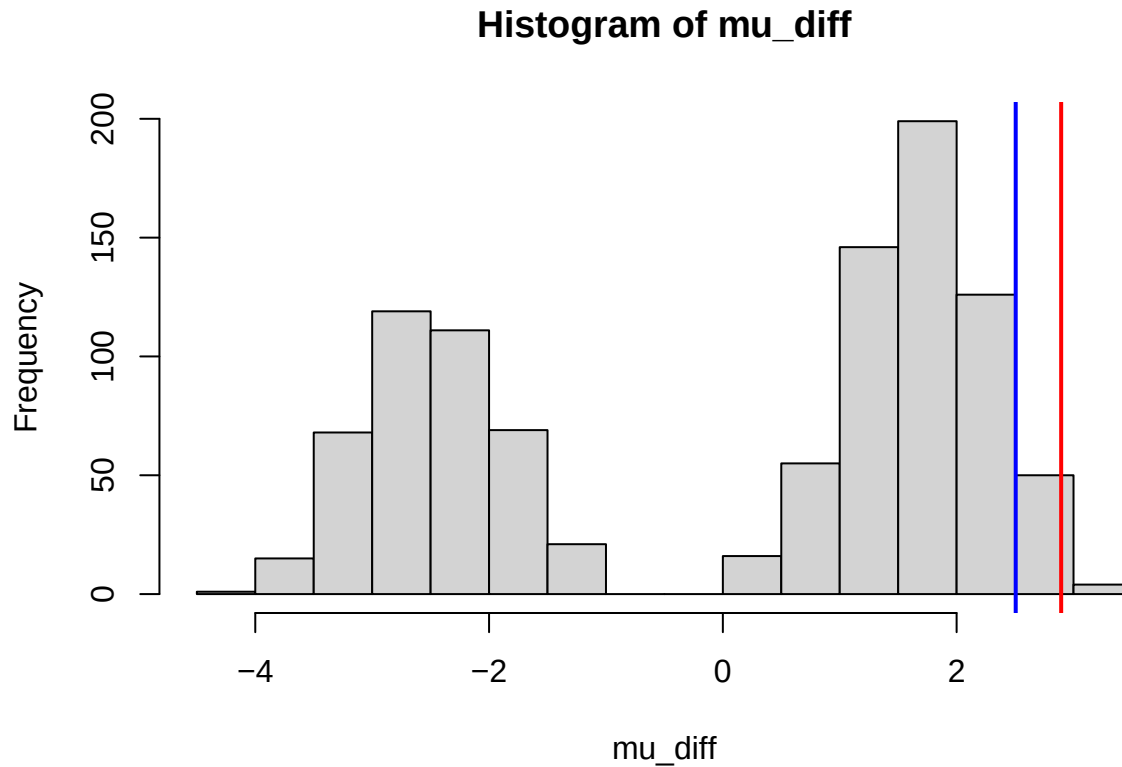
c

BCa works better when the distribution of our parameter is skewed which we can see in the above. The BCa interval is shifted to the right compared to the percentile bootstrap we calculated and the interval assuming

normality. Normality is probably a strong assumption here. The travel-time is around 15-23 minutes.

d

A reasonable estimate is the difference in mean of each group. We break the signal between the travel time and if it is in the afternoon or evening. If we do that, what variance in group mean difference can we expect given how varied our data is? In the plot below we see that less than 5% of differences are as large as our observed difference leading us to conclude that the traveltime is larger in the evenings.



Observed group difference: 2.894444

Appendix

```
inv_cdf <- function(x, mu = 0) {  
  c <- log(log(2))  
  mu + c - log(-log(x))  
}
```

```
n <- 13
```

```
set.seed(1234)
```

```
U <- runif(n)
```

```

X <- inv_cdf(U, mu = 5)
hist(X)

std_error <- function(p, n = 10) {
  sqrt(p*(1-p)/n)
}

par(mfrow = c(2, 2))

curve(std_error(x, 10), c(0, 1))
curve(std_error(x, 25), c(0, 1))
curve(std_error(x, 50), c(0, 1))
curve(std_error(x, 1000), c(0, 1))

curve(std_error(x, 5000), c(0, 1))

mu_vec <- seq(0, 2, by = 0.1)
rep <- 5000
powers <- c()

set.seed(1234)

i <- 1
for (mu in mu_vec) {

  rejected_nulls <- 0

  for (j in 1:rep) {
    U <- runif(n)
    X <- inv_cdf(U, mu)
    test <- SIGN.test(X, alternative = "greater")

    # If the lower confidence limit is larger than 0 we reject the null.
    rejected_nulls <- rejected_nulls + (test$conf.int[1] > 0)
  }

  power <- rejected_nulls/rep
  powers <- c(powers, power)
}

plot(mu_vec, powers, t = "l")

data <- c(16.5, 14.6, 15.0, 13.9, 15.4, 29.7, 14.8, 15.3, 17.0)

```

```

mu <- c()
B <- 10000

set.seed(123)
for (i in 1:B) {
  bootstrap_sample <- sample(data, length(data), replace = T)
  mu <- c(mu, mean(bootstrap_sample))
}

hist(mu)
abline(v = mean(data), col = "red", lwd = 2)
abline(v = quantile(mu, c(0.025, 0.975)), col = "blue", lwd = 2)

cat("CI: ", quantile(mu, c(0.025, 0.975)))

boot_mean <- function(data, i) {
  mean(data[i])
}

set.seed(123)
cb <- boot(data, boot_mean, R=10000)
boot.ci(cb, type="bca")

# assuming normally distributed
t_test <- t.test(data)
t_test$conf.int

data2 <- c(15.0, 13.0, 12.9, 14.4, 15.1, 13.7)

df <- as.data.frame(matrix(0, length(data) + length(data2), 2))
colnames(df) <- c("data", "labels")
df$data <- c(data, data2)
df$labels <- c(rep(1, length(data)), rep(2, length(data2)))

index <- df$labels == 1
true_mu_diff <- mean(df$data[index]) - mean(df$data[!index])

B <- 1000

mu_diff <- c()
for (i in 1:B) {
  df$labels <- sample(df$labels)
  index <- df$labels == 1
  mu_diff <- c(mu_diff, mean(df$data[index]) - mean(df$data[!index]))
}

hist(mu_diff)

```



```
abline(v = true_mu_diff, col = "red", lwd = 2)
abline(v = quantile(mu_diff, c(0.95)), col = "blue", lwd = 2)

cat("Observed group difference: ", true_mu_diff)
```